

Graduate Institute of Artificial Intelligence

Master's Thesis Oral Defense

Span2Type: Open World Named Entity Recognition with Automatic Entity Naming

Presenter: **Nguyen Anh Khoa**

Student ID: **M1261026**

Advisor: **Hsien-Tsung CHANG 張賢宗**

Date: **2026/04/28**

Outline

- Introduction
- Challenges
- Related Work
- Methodology
- Experiments & Results
- Conclusion

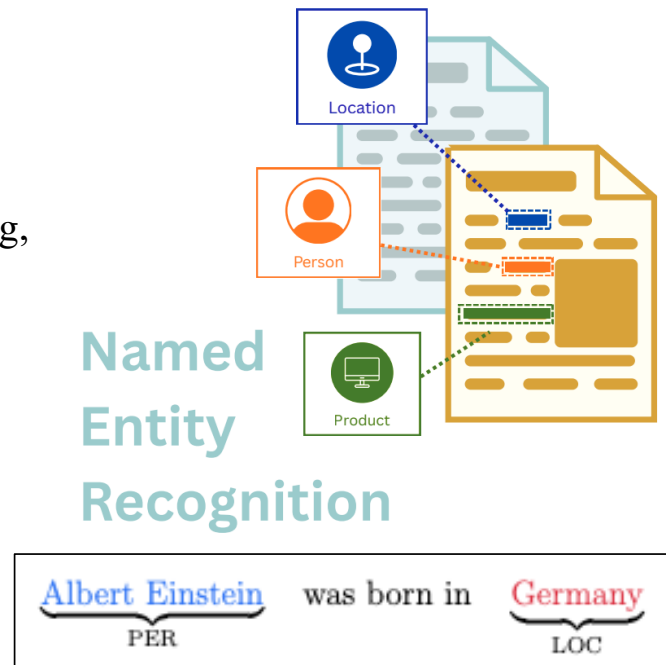
Introduction

From Traditional Named Entity Recognition to Open-World NER

- ❑ **Named Entity Recognition (NER)** is a core NLP task that:
 - **Detects** entity spans in text
 - **Classifies** them into semantic categories
 - BERT/RoBERTa: near-human F1 on CoNLL-2003
- ❑ **Application:** information extraction, knowledge graphs, text mining, question answering, document understanding
- ❑ **Traditional NER works well when:**
 - Entity types are **predefined**
 - Labeled training data is **available**
 - Source and target domains are similar

The hidden assumption

Type inventory is fixed at training time.
Labeled data must exist for every type.



Introduction

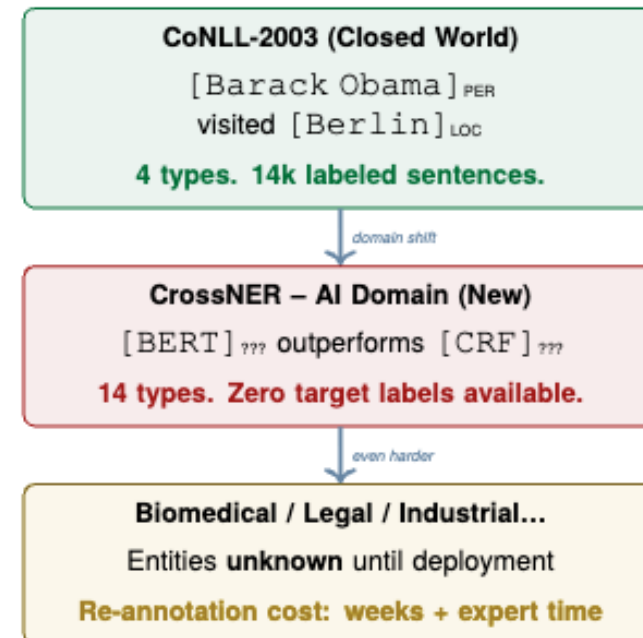
From Traditional Named Entity Recognition to Open-World NER

❑ In Practice, these assumptions often fail

- New domains appear frequently
- Entity types are unknown or evolving
- Annotation is expensive and slow
- Domain shift reduces model performance

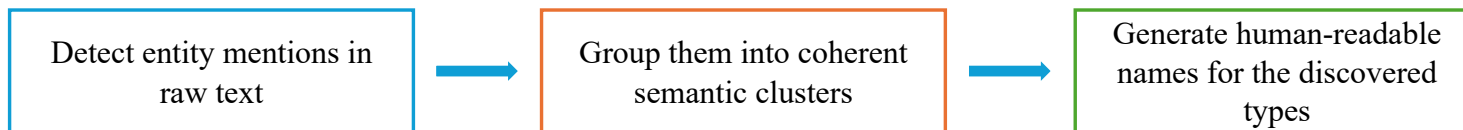
❑ Examples of new entity types

- Biomedical: drug, protein, disease
- Scientific: algorithm, benchmark, method
- Legal/political: law, treaty, office



Introduction – Open-World NER

- ❑ Open-world NER removes the closed-world assumption.
- ❑ The system is **not** given:
 - A predefined target-domain types
 - Labeled target-domain training data
- ❑ **Instead, it must:**



- ❑ **Why it is harder than traditional NER**
 - Mention boundaries must still be found accurately
 - Number of types is unknown
 - Clusters need interpretation
 - Discovered types may not perfectly match gold ontology labels

Challenges

- ❑ **Domain shift:** source-trained detectors may not generalize to new domains
- ❑ **Unknown types:** target-domain entity categories are not predefined
- ❑ **Interpretability:** discovered clusters need human-readable names
- ❑ **Evaluate** the Open-Vocabulary Type Named

Goal: Build a system that can detect, cluster, and name entities from unlabeled target-domain text.

Challenges

- ❑ **Domain shift:** source-trained detectors may not generalize to new domains
- ❑ Three complementary views exist:
 - **BIO** (global context)
 - **Start-End** (explicit boundary positions)
 - **Tie-Break** (adjacent co-membership)
- ❑ **Unknown types:** target-domain entity categories are not predefined

Source (newswire):

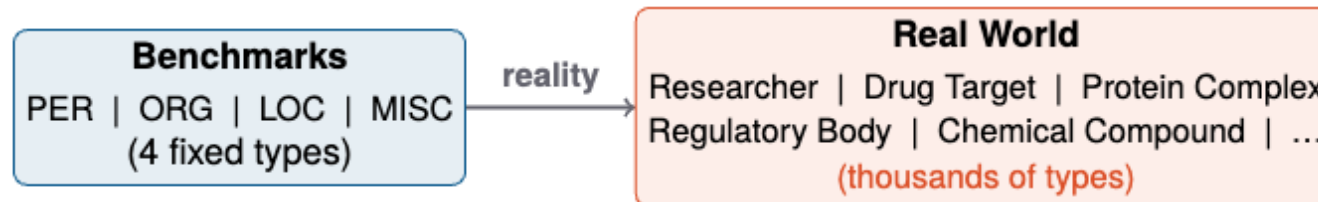
``Germany defeated Brazil 2-1.``

✓ Simple nouns – BIO handles fine

Target (CrossNER-AI):

``We apply **support vector machine**
to image classification.``

× 3-token span – BIO often splits it



Challenges

- ❑ **Interpretability:** discovered clusters need human-readable names
 - Transforms cluster assignments into a usable ontology

"Albert Einstein" → Cluster 3
"Marie Curie" → Cluster 3
"Watson" → Cluster 7

What is Cluster 3?
A domain expert cannot validate this.



Cluster 3 → "Physicist"
Cluster 7 → "Fictional Character"

Validatable.
Integrable into knowledge bases.
Usable in downstream pipelines.



- ❑ **Evaluate the Open-Vocabulary Type Named**
 - Open-world NER generates open-vocabulary names (e.g., synonyms or compound phrases).
 - Exact-match evaluation fails on valid variations (e.g., "musician" vs. "musicalartist").
 - Human evaluation is difficult to scale across domains and clusters, and is costly and time-consuming.

Gold: "musicalartist"
Predicted: "musician"
Exact-match: WRONG (0)
Reality: semantically correct

Gold: "researcher"
Predicted: "physicist"
Exact-match: WRONG (0)
Reality: close hyponym

Related Work

Traditional Supervised NER

- ❑ Early approaches: **CRF-based sequence labeling**
- ❑ Later approaches: **BiLSTM-CRF**. Recent approaches: **BERT / RoBERTa-based tagging**
- ❑ Strong performance on benchmarks such as **CoNLL-2003**
- ❑ Main limitation:
 - assumes a **fixed label set**
 - requires **labeled training data**

Open-World / Unsupervised NER

- ❑ Removes the closed-world assumption
- ❑ The model must:
 - Detect mentions from unlabeled text
 - Represent entity semantics
 - Cluster mentions into latent types
- ❑ Representative systems: **UniNER / UniversalNER ,ChatIE, OWNER**

Cross-Domain / Low-Resource NER

- ❑ Aims to reduce annotation cost
- ❑ Uses: Transfer learning, Domain adaptation, Few-shot learning
- ❑ Improves generalization to new domains
- ❑ But still usually assumes:
 - Predefined entity types
 - At least some target-domain supervision

Common Limitation across all eras

- Require predefined type inventory
- Need at least a few labeled examples per type
- Cannot discover new entity categories

Open-World NER remove all constraints

- No predefined types
- No target-domain annotations
- Discovers types automatically via clustering

Related Work

Traditional Supervised NER

- ❑ Early approaches: **CRF-based sequence labeling**
- ❑ Later approaches: **BiLSTM-CRF**. Recent approaches: **BERT / RoBERTa-based tagging**
- ❑ Strong performance on benchmarks such as **CoNLL-2003**
- ❑ Main limitation:
 - assumes a **fixed label set**
 - requires **labeled training data**

Open-World / Unsupervised NER

- ❑ Removes the closed-world assumption
- ❑ The model must:
 - Detect mentions from unlabeled text
 - Represent entity semantics
 - Cluster mentions into latent types
- ❑ Representative systems: **UniNER / UniversalNER ,ChatIE, OWNER**

Cross-Domain / Low-Resource NER

- ❑ Aims to reduce annotation cost
- ❑ Uses: Transfer learning, Domain adaptation, Few-shot learning
- ❑ Improves generalization to new domains
- ❑ But still usually assumes:
 - Predefined entity types
 - At least some target-domain supervision

Key baseline for this thesis

- ❑ **OWNER**
 - BIO-based mention detector
 - fill-in-the-blank entity encoding
 - BIC-guided k-means clustering
 - MLM/LLM-based cluster naming

Related Work

Span Detection

❑ Sequence tagging (BIO):

- Standard: BERT/DeBERTa + single BIO head
- Error propagation: wrong B tag corrupts entire span
- Single boundary signal — easily confused cross-domain

❑ Span enumeration (PURE, Chen '21):

- Enumerate all $O(n^2)$ candidate spans explicitly
- No boundary cascading errors
- Weak cross-domain transfer — F1 35.8 avg on CrossNER

❑ Multi-view ensemble (DTrans, Zheng '24):

- Three complementary decoders trained jointly:
BIO + Start-End (SE) + Tie-Break (TB)
- Mean Teacher EMA enforces cross-view consistency
- Errors in one view cancelled by consensus of others
- F1 82.4 avg — strongest cross-domain detection

Related Work

Entity type generation:

- ❑ Supervised NER: human-crafted fixed inventory (PER, ORG, LOC...) — requires domain experts
- ❑ Few-shot NER (FewNERD, EpiMeta): type names still required at test time from the user
- ❑ OWNER '23: auto-discover types via clustering — first system to generate type list automatically
- ❑ Limitation: cluster names still coarse/redundant

Cluster naming — exemplar selection strategies:

- ❑ OWNER (MLM): uses all entities in a cluster — no sampling, but noisy & slow on large clusters
- ❑ OWNER (LLM) / Random: subsample n=16 randomly — cheap, unstable (misses semantic range)

OWNER: The State-of-the-Art Baseline (Wang et al., 2023)

❑ OWNER pipeline – 3 stages

1. Mention Detection:

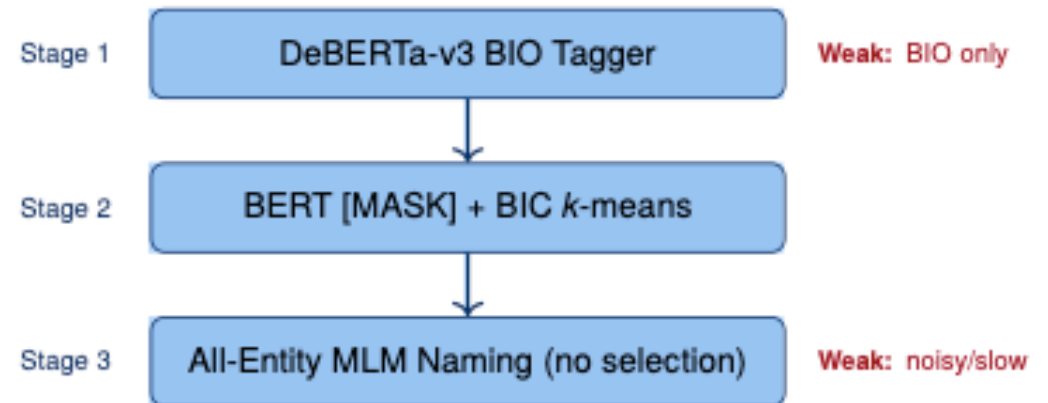
- *DeBERTa-v3-base* + single BIO head
trained on source domain

2. Entity Typing:

- BERT fill-in-the-blank template
- **K-means** with BIC-guided auto-k

3. Cluster Naming:

- All cluster entities — no selection
Single template + BERT MLM majority-vote / Llama3:8b



❑ Strong baseline: 49.4 avg AMI on CrossNER

Thesis Objectives

Overarching Contribution

- ❑ Build a complete framework: given an unannotated target corpus, automatically produce entity spans each assigned a human-readable type name — no predefined type list, no target-domain labels.

Technical Objectives:

- ❑ Integrate multi-view detection (BIO + SE + TB) into the OWNER pipeline
- ❑ Develop two-stage soft-prompt entity typing — better initialization, minimal forgetting
- ❑ Propose MMR-based exemplar selection for more specific, stable cluster naming

Key constraint :

- ❑ **Zero target-domain labels at any stage.**

Methodology

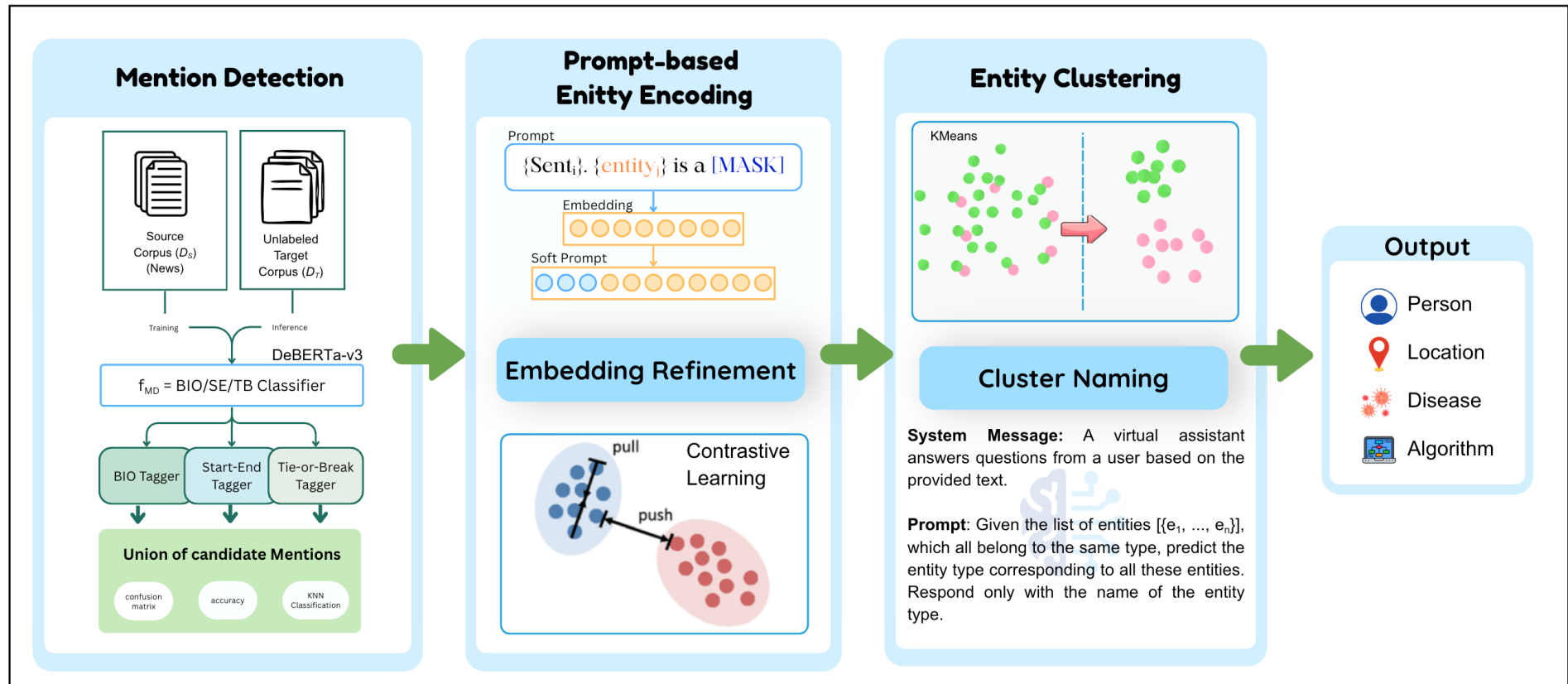


Figure 1: Overview of the Span2Type framework. The pipeline consists of three stages: (1) Mention Detection, (2) Entity Representation, (3) Entity Clustering and Cluster Naming

Methodology – Stage 1: Multi-view Entity Detection

Why single-view BIO is insufficient:

- ❑ Classifies each token as B/I/O in isolation
- ❑ Struggles with multi-token technical spans: support vector machine convolutional neural network
- ❑ One wrong token-tag → entire span lost

Three complementary views of a span:

- ❑ **BIO** — global sequential context
 - ❑ **Start-End (SE)** — explicit boundary positions
 - ❑ **Tie-Break (TB)** — pairwise token adjacency
- ➔ Union of three views ➔ Higher recall under domain shift

Example: “support vector machine”

tokens	support	vector	machine	achieved
BIO	B	I	I	O
SE start	1	0	0	0
SE end	0	0	1	0
TB	Tie	Tie	Break	

All three views agree on the same 3-token span.
Even if BIO mislabels 1 token, SE or TB recovers it

Methodology

Multi-view Architecture

Backbone: BERT-base-cased (12L, hidden $d = 768$)

Input $\mathbf{x} \rightarrow \text{encoder} \rightarrow \mathbf{H} \in \mathbb{R}^{T \times d}$

Three parallel heads:

BIO decoder:

$$p_t^{\text{BIO}} = \text{softmax}(\mathbf{W}_{\text{bio}} \mathbf{h}_t)$$

where $\mathbf{W}_{\text{bio}} \in \mathbb{R}^{3 \times d}$

Start-End decoder:

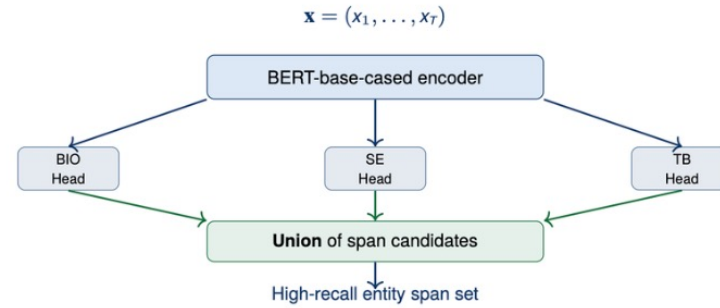
$$p_t^{\text{end}} = \text{softmax}(\mathbf{W}_{\text{end}}[\mathbf{h}_t; \hat{p}_t^{\text{start}}])$$

End conditions on predicted start $\Rightarrow$ boundary coherence

Tie-Break decoder:

$$p_t^{\text{tb}} = \text{softmax}(\mathbf{W}_{\text{tb}}(\mathbf{h}_t + \mathbf{h}_{t+1}))$$

Total loss: $L_{ED} = L_{BIO} + L_{SE} + L_{TB}$



Methodology – Stage 2: Entity Typing – Entity Representation

Template construction:

$\underbrace{[\text{CLS}] x_1 \cdots x_T}_{\text{sentence}} \underbrace{m}_{\text{entity}} \text{ is a } \underbrace{[\text{MASK}]}_{\text{type}} . [\text{SEP}]$

- Repeating the span m gives BERT an explicit type cue
- Leverages BERT's cloze pretraining knowledge

Entity embedding:

$$\mathbf{e}_s = \text{Dropout}(\mathbf{h}_{[\text{MASK}]}) \in \mathbb{R}^d$$

Training: Batch Triplet Margin Loss

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} \max(0, d_{ij} - d_{ik} + \gamma)$$

Concrete Example

Sentence: *Albert Einstein was born in Germany*

Span: *Albert Einstein*

Template:

[CLS] Albert Einstein was born
in Germany **Albert Einstein**
is a [MASK] . [SEP]

BERT predicts: *physicist, scientist*

$\mathbf{e}_s = [\text{MASK}]$ hidden state

→ rich type-semantic representation

Methodology – Stage 2: Entity Typing – Soft Prompt Variant

Soft prompt (P-tuning / Prefix-tuning):

A fixed prompt uses manually designed text

Soft prompt tuning learns the prompt automatically

Trainable virtual tokens are added before the input

These tokens are updated by gradient descent

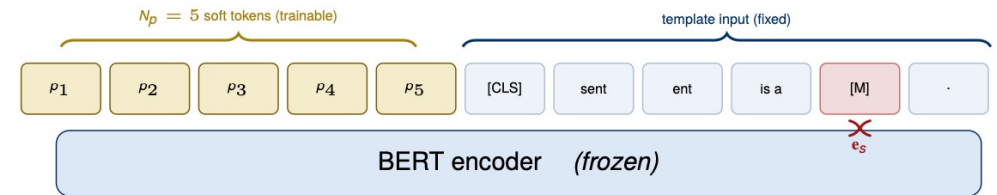
This helps the model adapt the prompt to the task without changing much of the backbone model

$N_p = 5$ trainable tokens $\mathbf{P} \in \mathbb{R}^{N_p \times d}$

Prepended to template embeddings:

$$\text{inputs} = \left[\underbrace{\mathbf{P}}_{\text{trainable}} ; \underbrace{\mathbf{E}_x}_{\text{frozen}} \right] \in \mathbb{R}^{(N_p+L) \times d}$$

BERT backbone: **strictly frozen**



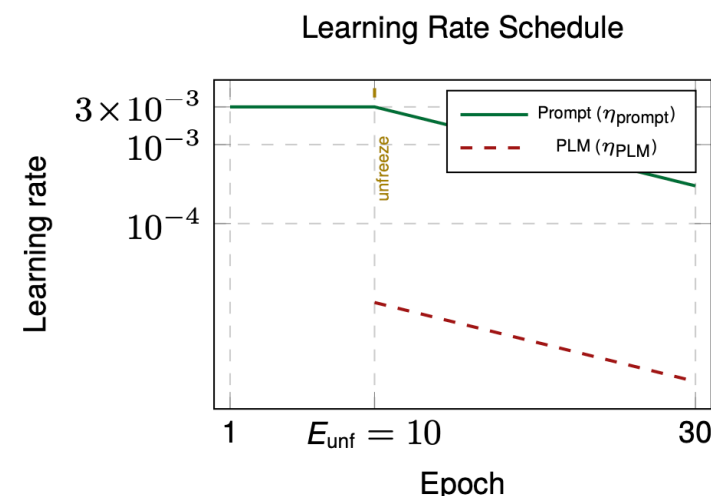
Methodology – Stage 2: Entity Typing – Two-Stage Soft Prompt Fine-Tuning

❑ Problem:

- Soft prompt only → limited expressiveness
- Full fine-tuning from epoch 1 → catastrophic forgetting

❑ Solution: 2 stage schedule at $E_{\text{unfreeze}}=10$

	Stage 1 Epochs 1–9	Stage 2 Epochs 10–30
Trainable	prompt only	prompt + BERT
LR (prompt)	3×10^{-3}	3×10^{-3}
LR (BERT)	frozen	1×10^{-5}
PLM frozen	✓	✗



Why this works

Stage 1: prompt warms up rapidly (3×10^{-3}) without disturbing pretrained BERT.

Stage 2: BERT adapts at 1×10^{-5} — 300× smaller — preserving pretrained representations.

Methodology – Stage 2: Entity Typing – Find number of cluster

Challenge: target domain has *unknown* k

AutoKmeans: search $k \in [2, 30]$, step 2

Bayesian Information Criterion:

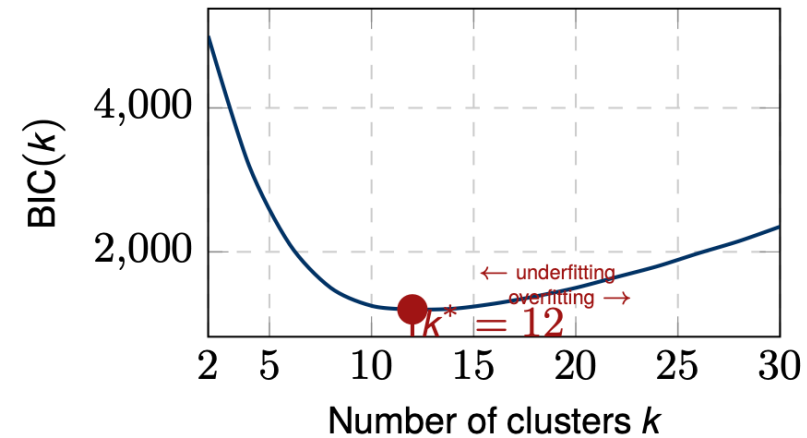
$$\text{BIC}(k) = n \cdot \log\left(\frac{\mathcal{I}(k)}{n}\right) + \log(n) \cdot k$$

- $\mathcal{I}(k)$: inertia (sum of squared distances)
- First term: fit quality — decreases with k
- Second term: complexity penalty — grows with k

$$k^* = \arg \min_k \text{BIC}(k)$$

- k-means++ init, 10 random restarts per k
- **Retained unchanged from OWNER**

BIC curve (schematic)



After k^* is selected, embeddings $\{\mathbf{e}_s\}$ are assigned cluster labels $c_s \in \{1, \dots, k^*\}$ for all target-domain spans.

Methodology – Stage 3: Cluster Naming

Problem with centroid-nearest selection:

- 16 exemplars all near centroid → redundant
- Single surface form repeated → unstable naming

Maximal Marginal Relevance (MMR):

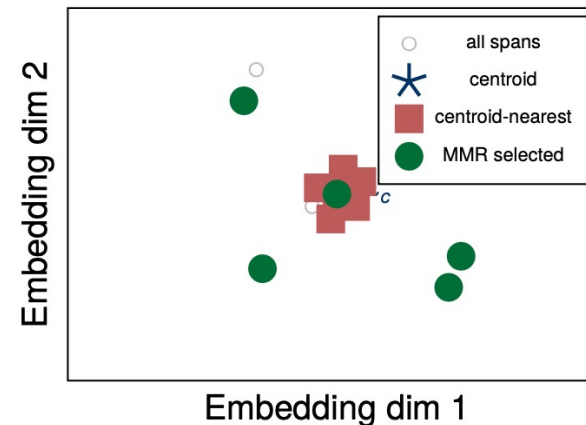
$$\text{MMR}(s) = \lambda \cdot \cos(\mathbf{e}_s, \boldsymbol{\mu}_c) - (1-\lambda) \cdot \max_{r \in \mathcal{R}} \cos(\mathbf{e}_s, \mathbf{e}_r)$$

- First term: **relevance** to cluster centroid $\boldsymbol{\mu}_c$
- Second term: **diversity** from already-selected $\mathcal{R}$
- $\lambda = 0.7$: slightly favours relevance over diversity
- Iterate until $|\mathcal{R}| = K_e = 16$

Effect

Exemplars cover the **full semantic spread** of the cluster, not just the centroid cloud

Centroid vs. MMR Selection



MMR (green) covers the cluster periphery; centroid-nearest (red) is redundant.

MMR (green) covers the

Methodology – Stage 3: Cluster Naming

Masked Language Model (BERT)

- ❑ Each selected sentence $x^{(r)}$ containing entity mention $m^{(r)}$ is inserted into a naming template

In this sentence, $m^{(r)}$ is a [MASK]. Sentence: $x^{(r)}$.

- ❑ Post-filter: remove subwords, stopwords, punctuation.
- ❑ Output: single token (conference, band, protein)
- ❑ Cannot express compound types: musical artist, literary genre

Large Language Model (Llama3:8b)

System Message: A virtual assistant answers questions from a user based on the provided text.

Prompt: Given the list of entities $[\{e_1, \dots, e_n\}]$, which all belong to the same type, predict the entity type corresponding to all these entities. Respond only with the name of the entity type.

- ❑ Temperature = 0 (deterministic).
Post-process: ≤ 3 words, lowercase, strip fillers.
- ❑ Output: multi-word labels- political party, research institute
- ❑ Immune to MLM hallucination artefact

Experimental Setup - Dataset

Source training data:

- **CoNLL-2003**: English newswire, 4 types, 14k sentences
- **Pile-NER**: diverse web corpus, ~ 13 k types, 50k sentences

Target evaluation (CrossNER):

Domain	#Test	#GT Types
AI	431	14
Literature	416	12
Music	465	13
Politics	650	9
Science	543	17

Gold labels used **only for evaluation** — never seen during training or clustering.

Key hyperparameters:

Component	Setting
Detector backbone	BERT-base-cased
Detector LR	1×10^{-5} , 50 epochs
EMA decay	$\alpha = 0.995$
Typing backbone	BERT-base-uncased
Full FT LR	2×10^{-5} , 30 epochs
Soft prompt N_p	5 tokens
Stage 1 LR	3×10^{-3} , 10 epochs
Stage 2 LR	1×10^{-5} , 20 epochs
Triplet margin	$\gamma = 1.0$
Clustering range	$k \in [2, 30]$, step 2
k-means restarts	10 per k
MMR exemplars	$K_e = 16$, $\lambda = 0.7$
LLM	Llama-3 via Ollama

Experimental Setup – Evaluate on each stage

Stage 1- Detection

- Span F1
- Measures whether the Multiview transfer better than BIO

Stage 2 - Typing

- AMI (Adjusted mutual information)
- Robust to permutations, over-segment, over-merging
- 0- random, 1-perfect

Stage 3 - Naming

- BERTScore-F1(RoBERTa-large)
- Semantic similarity of predicted vs gold type name

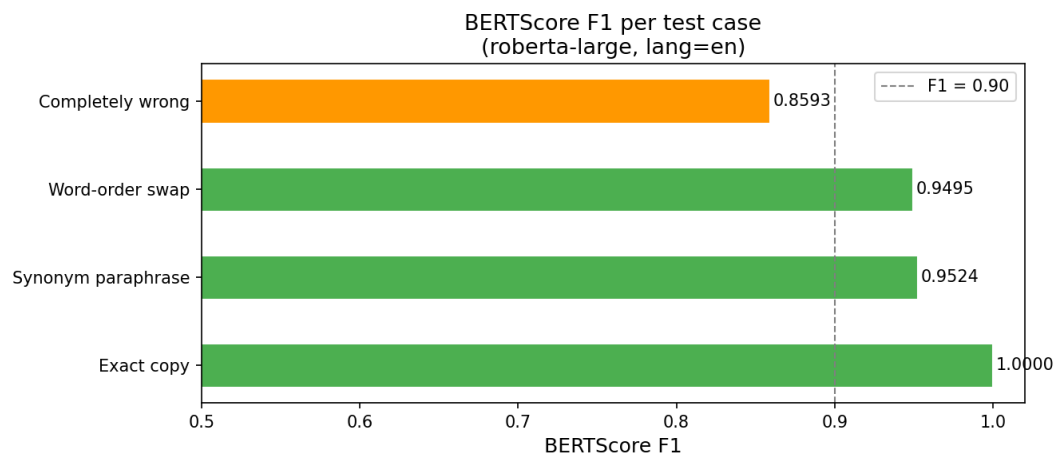
AMI measures how well predicted clusters match gold entity groups, while correcting for agreement that could happen by chance.

BERTScore interpretation:

Meaningful range is 0.86–1.0 (even a wrong label scores ≈ 0.8 due to shared embedding space).

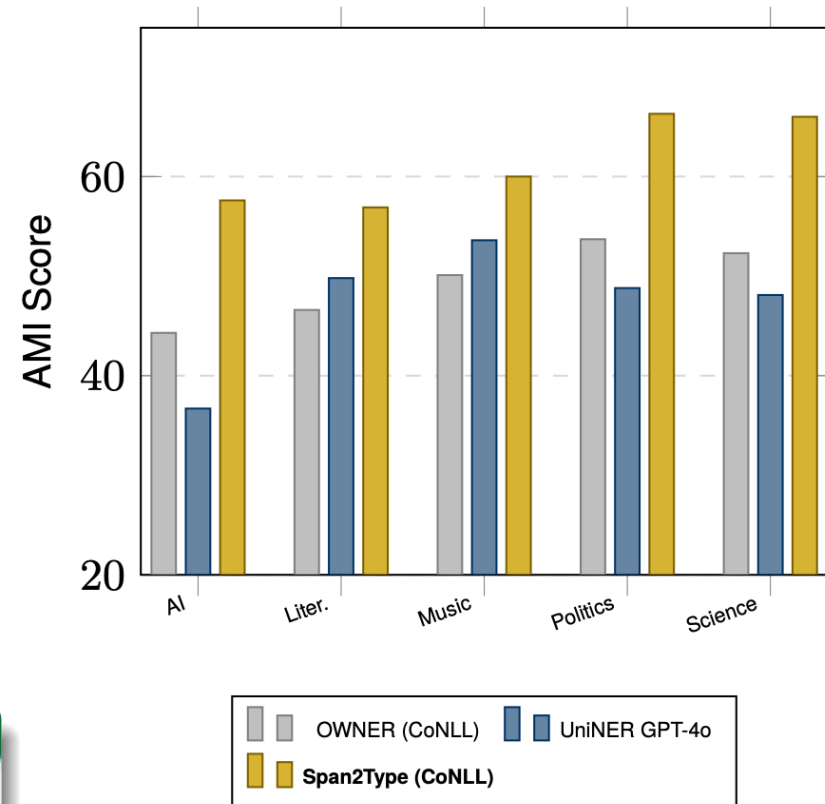
Exact copy = 1.0

Synonym paraphrase ≈ 0.95 Related concept ≈ 0.90



Main Results: AMI on CrossNER

Method	AI	Lit	Mus	Pol	Sci	Avg
<i>Unsupervised Baselines</i>						
ChatIE (GPT-4o)	21.8	37.5	43.7	37.2	35.0	35.0
UniNER	33.5	42.8	48.1	40.3	43.7	41.7
UniNER (GPT-4o)	36.7	49.8	53.6	48.8	48.1	47.4
OWNER (CoNLL)	44.3	46.6	50.1	53.7	52.3	49.4
OWNER (Pile)	39.4	49.5	52.5	48.5	50.9	48.2
<i>Proposed</i>						
Span2Type (Pile)	50.0	52.8	56.2	55.8	62.3	55.4
Span2Type (CoNLL)	57.6	56.9	60.0	66.3	66.0	61.4



Key numbers

vs. OWNER: **+12.0 pts** avg AMI

vs. best LLM baseline (UniNER GPT-4o): **+14.0 pts**

Results – Entity Detection F1: Multi-view vs Single-View

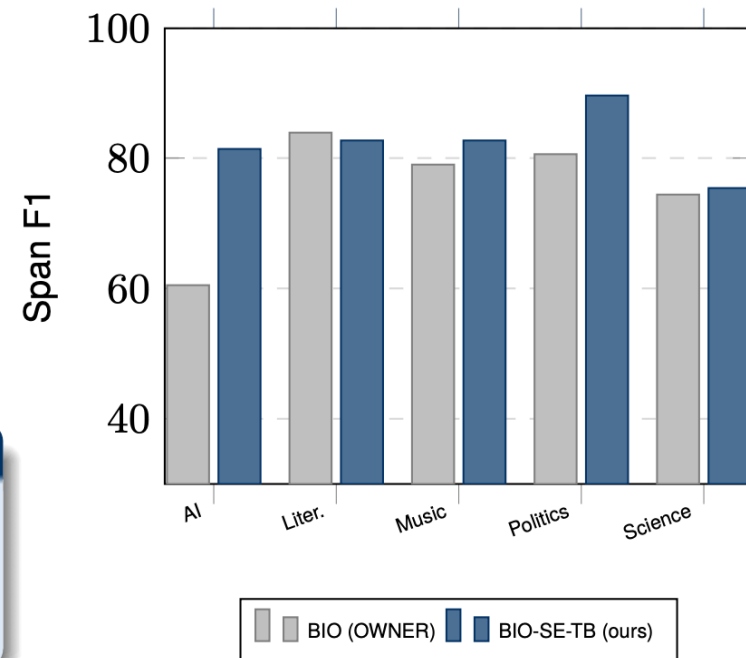
Model	AI	Lit	Mus	Pol	Sci	Avg
PURE	39.8	37.1	33.8	32.4	35.7	35.8
SpanProto	54.1	62.9	59.6	68.7	59.7	61.0
WL-Coref	57.4	68.3	66.9	72.1	63.4	65.6
BIO (OWNER)	60.5	83.9	79.0	80.6	74.4	75.7
BIO-SE-TB	81.4	82.7	82.7	89.6	75.4	82.4

Average gain

OWNER BIO: 75.68

DTrans BIO-SE-TB: **82.35**

Improvement: **+6.67 F1 points**



Literature

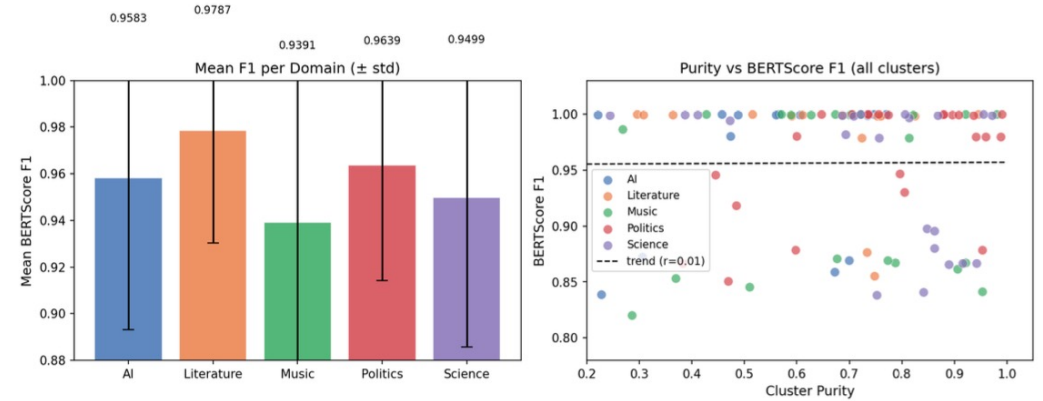
regression: -1.22 — literary entities
(character names, book titles) follow predictable
proper-noun patterns; SE/TB add noise, not signal.

Results – Clustering Naming: BERTScore-F1 Results

Method	AI	Lit	Mus	Pol	Sci	Avg
#Clusters	14	14	22	24	22	–
#GT Types	14	12	13	9	17	–
OWNER MLM	.989	.974	.958	.964	.945	.966
OWNER LLM	.929	.954	.957	.953	.926	.944
S2T Centroid	.904	.887	.895	.883	.894	.892
S2T MLM+MMR	.900	.889	.911	.887	.893	.896
S2T LLM+MMR	.958	.979	.939	.964	.950	.956

Budget-controlled comparison ($n = 16$, same LLM):

- Span2Type-LLM (MMR): 0.956
- OWNER (LLM random): 0.944
- **MMR gain: +0.012 avg**



Left: mean BERTScore-F1 $\pm$ std per domain.

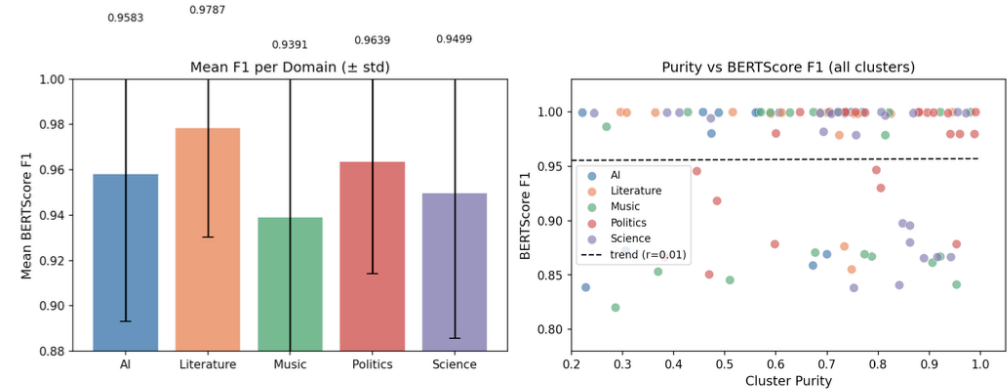
Right: cluster purity vs. BERTScore-F1;
 $r = 0.01$ — naming quality is **independent**
of cluster purity. LLM recovers semantics
even from impure clusters.

Results – Clustering Naming: BERTScore-F1 Results

Method	AI	Lit	Mus	Pol	Sci	Avg
#Clusters	14	14	22	24	22	—
#GT Types	14	12	13	9	17	—
OWNER MLM	.900	.891	.906	.892	.894	.897
OWNER LLM (rand.)	—	.923	—	—	—	.908
S2T Centroid	—	—	—	—	—	.893
S2T MLM+MMR	—	—	—	—	—	.896
S2T LLM+MMR	.922	.913	.943	.928	.912	.924

Budget-controlled comparison ($n = 16$, same LLM):

- Span2Type-LLM (MMR): **0.924**
- OWNER (LLM random): 0.908
- **MMR gain: +0.016 avg**
- Lit exception: OWNER LLM (0.923) > ours (0.913)



Left: mean BERTScore-F1 $\pm$ std per domain.

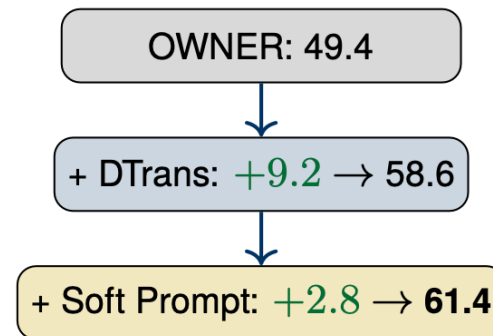
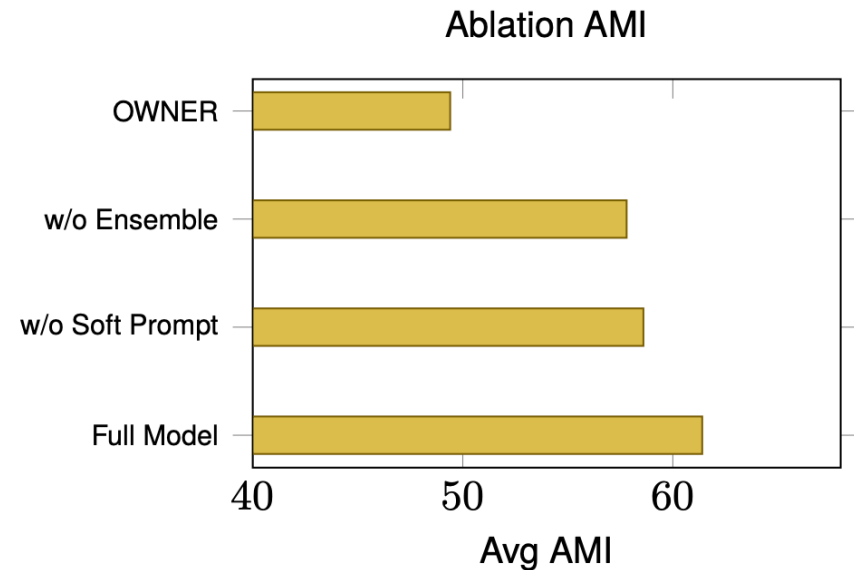
Right: cluster purity vs. BERTScore-F1;
 $r = 0.01$ — naming quality is **independent**
of cluster purity. LLM recovers semantics
even from impure clusters.

Ablation Study: Contribution of Each Component

Model	AI	Lit	Mus	Pol	Sci	Avg
OWNER (baseline)	44.3	46.6	50.1	53.7	52.3	49.4
w/o Soft Prompt	52.3	56.3	57.5	63.9	62.8	58.6
w/o Ensemble MD	49.3	53.0	58.8	64.1	63.8	57.8
Full Model	57.6	56.9	60.0	66.3	66.0	61.4

Conclusions:

- 1 DTrans multi-view: **+9.2** — largest single gain
- 2 Soft prompt: **+2.8** — on top of DTrans
- 3 Both contribute independently: removing either causes a drop



Discussion: Key Findings & Insights

❑ Why CoNLL outperforms Pile-NER?

- CrossNER is Wikipedia-style formal text
- CoNLL-2003 newswire $\approx$ similar syntax and capitalization conventions
- Pile-NER: broader vocab but noisier, colloquial
- Source-target stylistic alignment > vocabulary breadth

❑ Cascade effect: detection $\rightarrow$ AMI

- Span F1 gain: +6.67
- AMI gain: +12.0
- End-to-end gain **exceeds** raw detection gain
- Hypothesis: better spans $\rightarrow$ cleaner [MASK] representations $\rightarrow$ more coherent clusters

❑ When does MMR hurt?

- Music domain: OWNER random (0.957) > MMR (0.939)
- Music clusters are tight & homogeneous: band, genre, album — unambiguous
- MMR's diversity pressure selects peripheral outliers $\rightarrow$ noise
- Rule: use MMR for heterogeneous clusters; centroid-nearest for tight clusters

❑ Naming $\perp$ clustering quality

- Pearson $r = 0.01$ between cluster purity and BERTScore
- LLM can identify dominant type even from impure clusters
- Improving k estimation benefits AMI, not naming quality

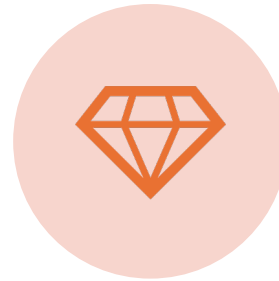
Summary of Contributions

- ❑ Multi-view entity detection
 - DTrans: BIO + SE + TB ➔ Span F1: 75.68 → 82.35 (+6.67)
 - Largest gain in AI domain: +20.85
- ❑ Two-stage soft-prompt entity typing
 - Stage 1: prompt warmup (3×10^{-3})
 - Stage 2: full model (1×10^{-5})
 - AMI contribution: +2.8 points
- ❑ MMR-based cluster naming
 - K = 16 diverse exemplars, $\lambda = 0.7$
 - LLM backend: 0.956 avg BERTScore-F1
 - +0.012 over random selection (budget-controlled)

Limitation



Detector still requires labeled source domain (newswire → clinical text: large gap)



BIC overestimates k for imbalanced domains (Politics: 24 predicted vs. 9 gold types)



MLM names: limited to single token
LLM names: requires locally hosted model



MMR counterproductive for tight, homogeneous clusters (Music)

Conclusion

- ❑ Proposed **Span2Type**, a three-stage pipeline for:
 - Entity detection
 - Entity typing
 - Automatic cluster naming
- ❑ **Main achievements**
 - **Multi-view entity detection** improved cross-domain span detection
 - **Prompt-based entity typing** produced more discriminative entity representations
 - **MMR-based cluster naming** generated more interpretable type labels
 - The full system outperformed the **OWNER** baseline in clustering performance on CrossNER
- ❑ **Final message**
 - No predefined target-domain type inventory
 - No target-domain annotations
 - More robust and interpretable open-world NER pipeline

Future Directions:

- ❑ Domain-adaptive pretraining or self-training to bridge larger source-target gaps
- ❑ Hierarchical clustering to produce entity type ontologies rather than flat partitions
- ❑ MMR at training time — diverse in-context retrieval during entity typing
- ❑ Evaluate on biomedical & legal domains where open-world NER is most needed

Q & A



**Thank You for
Your Attention!**

Appendix A - Naming Examples (Qualitative)

Domain	GT Type	LLM Name	GT Type	MLM Name
AI	researcher	computer scientist	conference	conference
Literature	writer	novelist	book	novel
Music	musicalartist	musician	band	band
Politics	politician	political leader	politicalparty	democrat
Science	protein	biological protein	scientist	physicist

Failure patterns:

- **Specificity mismatch:** LLM predicts hyponym (*actor* for *person*)
- **Compound label penalty:** `musicalartist` vs. *musical artist* — semantically correct but tokenization mismatch reduces BERTScore
- **MLM hallucination:** *vegetarian* — BERT defaults to high-frequency unrelated token for OOD context. LLM backend is immune.

Appendix B – Literature Domain Per Cluster Naming

Cluster	GT Type	LLM Name	BERTScore	
C1	person	author	0.994	✓
C2	book	novel	0.998	✓
C3	location	theatre	0.979	~
C4	writer	poet	0.992	✓
C5	book	mythology	0.996	✓
C6	literarygenre	literary genre	0.993	✓
C7	award	literary award	0.998	✓
C8	event	film festival	0.877	✗
C13	literarygenre	literary work	0.855	✗

Literature = highest domain mean BERTScore-F1 = **0.979**

11 of 14 clusters score ≥ 0.99 (green).

C8 failure: specificity mismatch (*film festival* vs. broad event).

C13 failure: compound string `literarygenre` penalizes correct multi-word name.

Appendix C – Why LLM baselines Underperform

LLM-prompting baselines (UniNER, ChatIE) ask the LLM to identify AND type entities in one pass.

Three structural weaknesses vs. Span2Type:

- ❑ No explicit span detector: zero-shot extraction limits recall; the LLM’s span boundary decisions are not trained discriminatively on entity data.
 - ❑ Inconsistent type strings: open-vocabulary output from the LLM produces “a person working in politics” and “politician” for the same type — clustering these together is impossible and AMI scores drop.
 - ❑ Structural mismatch with AMI: AMI measures agreement between predicted and gold span partitions; LLM-extracted spans differ structurally from gold annotation boundaries, creating systematic offset even when semantics are correct.
- ➔ Span2Type separates span detection (discriminative), embedding (template), and clustering (unsupervised) into well-defined, optimized stages — each addressing one specific failure mode.